

Panel MS-AR(1) on SOE log real GDP per worker

Contents

1	Model	2
2	Common intercept and linear trend	3
3	Country-specific intercepts, common linear trend	4
4	Country-specific intercepts and linear trends	5
5	Country-specific CF filtered trend (25 years)	6
6	CF filtered trend, all regime means restricted to 0	7
7	CF filtered trend, common ρ	8

1 Model

Joint panel Markov-switching AR(1) around a trend, with three regimes:

$$y_{it} = a_i + g_i t + z_{it}, \quad (1)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it}) z_{it} + \sigma(s_{it}) \varepsilon_{it}, \quad (2)$$

$$s_{i,t+1} \sim \Pi(\cdot | s_{it}). \quad (3)$$

Countries are independent given shared Markov parameters; latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1} . σ switches with the regime. Unless noted, ρ is regime-specific and the median μ is pinned at 0 after ordering. The likelihood is a Hamilton filter per country, summed across the panel.

Trend assumptions

1. Common intercept a and common linear slope g (joint MLE).
2. Country intercepts a_i and common g (a_i profiled).
3. Country intercepts a_i and slopes g_i (both profiled).
4. Country-specific Christiano–Fitzgerald low-pass trend (cycle = oscillations of 2–100 quarters / 25 years), then MS-AR on the cycle.
5. Same CF trend, with every $\mu(s) = 0$.
6. Same CF trend, with one ρ common to all regimes.

Sections 4–6 are two-step: the trend is not re-estimated jointly with the cycle. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters only. A dashed entry is a pinned coefficient.

2 Common intercept and linear trend

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 17917.39. Common intercept $a = -4.0009$. Common slope $g = 0.0139$.

Table 1: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.1033	0.0000	0.7480
σ	0.0099 (0.0002)	0.0222	0.0692
ρ	0.9900	0.9900	0.9900
Π_0	0.9754 (0.0044)	0.0070	0.0177 (0.0013)
Π_1	0.0133 (0.0023)	0.9691 (0.0029)	0.0176 (0.0021)
Π_2	0.0812 (0.0068)	0.1445 (0.0109)	0.7743 (0.0112)

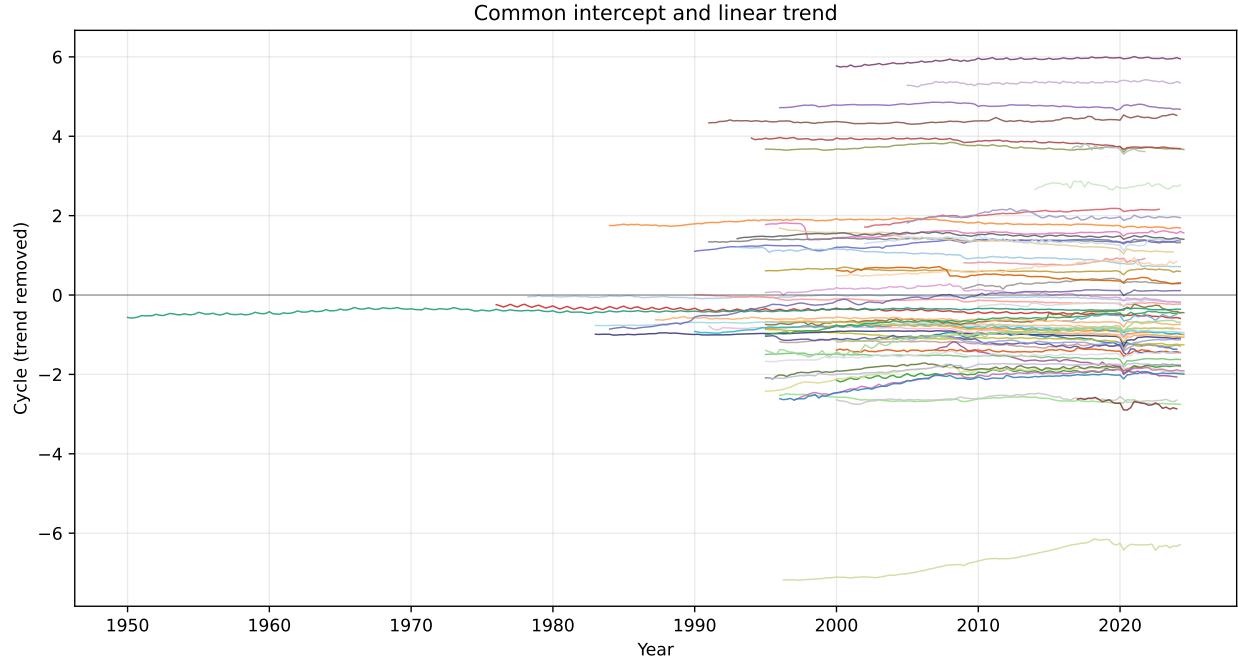


Figure 1: Country cycles after removing the section's trend.

3 Country-specific intercepts, common linear trend

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18079.16. Standard errors omitted. Country intercepts a_i : mean -3.863, min -10.356, max 2.154. Common slope $g = 0.0115$.

Table 2: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.1298	0.0000	0.2260
σ	0.0218	0.0682	0.0097
ρ	0.9900	0.8938	0.9892
$\Pi_0.$	0.9426	0.0200	0.0374
$\Pi_1.$	0.0932	0.7964	0.1104
$\Pi_2.$	0.0232	0.0106	0.9662

4 Country-specific intercepts and linear trends

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18357.24. Country intercepts a_i : mean -3.974, min -13.117, max 2.031. Country slopes g_i : mean 0.0140, min -0.0309, max 0.0547.

Table 3: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.0854 (0.0344)	0.0000 —	0.0341 (0.0005)
σ	0.0655	0.0094	0.0211
ρ	0.8490 (0.0677)	0.9751 (0.0056)	0.8981 (0.0098)
$\Pi_0.$	0.7770 (0.0263)	0.0740	0.1491
$\Pi_1.$	0.0204 (0.0016)	0.9717 (0.0032)	0.0080
$\Pi_2.$	0.0172 (0.0043)	0.0179 (0.0029)	0.9648 (0.0030)

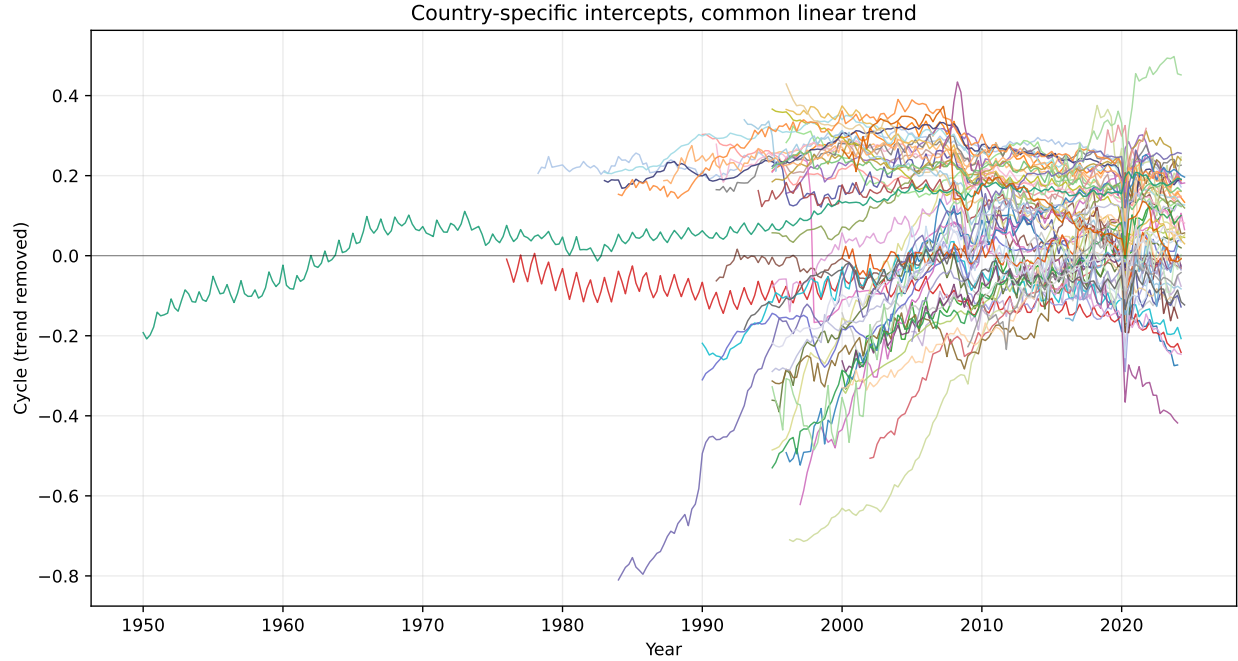


Figure 2: Country cycles after removing the section's trend.

5 Country-specific CF filtered trend (25 years)

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18637.48. Christiano–Fitzgerald low-pass trend: periods longer than 100 observations (25 years).

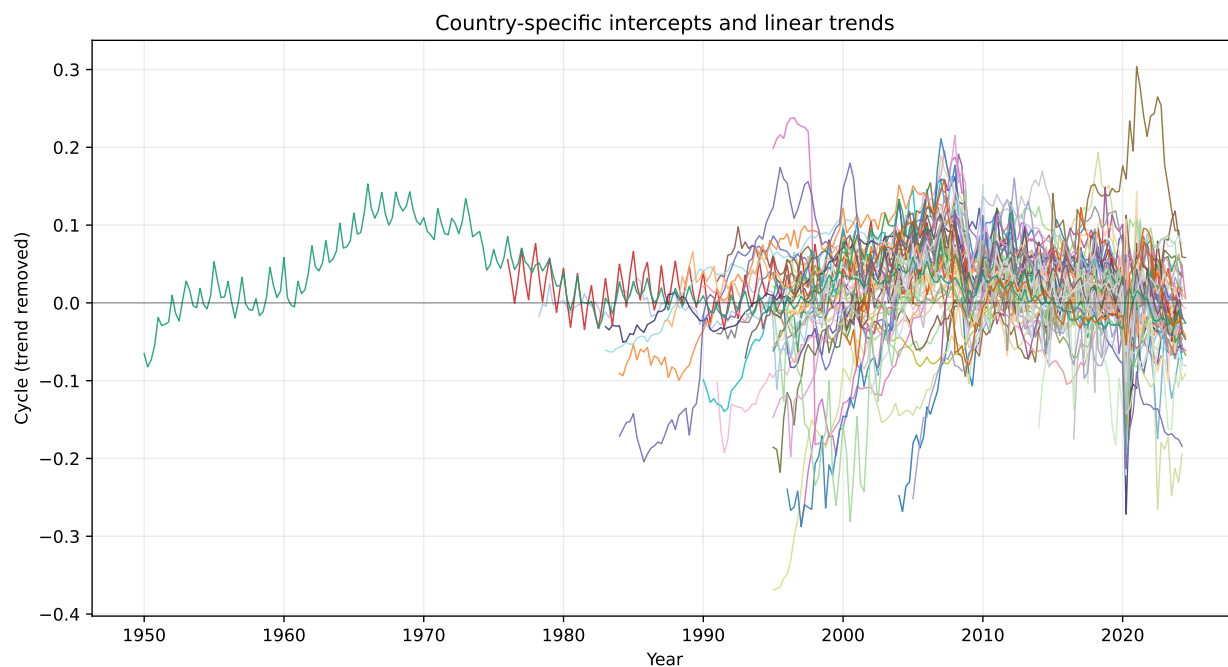


Figure 3: Country cycles after removing the section's trend.

6 CF filtered trend, all regime means restricted to 0

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18632.52. Christiano–Fitzgerald low-pass trend: periods longer than 100 observations (25 years).

Table 4: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.0173 (0.0054)	0.0000 —	0.0027 (0.0025)
σ	0.0582 (0.0026)	0.0091 (0.0002)	0.0204 (0.0005)
ρ	0.4857 (0.0470)	0.9242 (0.0072)	0.8484 (0.0115)
$\Pi_0.$	0.7641 (0.0317)	0.0682 (0.0193)	0.1676 (0.0325)
$\Pi_1.$	0.0191 (0.0033)	0.9725 (0.0034)	0.0085 (0.0031)
$\Pi_2.$	0.0208 (0.0044)	0.0164 (0.0040)	0.9628 (0.0055)

7 CF filtered trend, common ρ

SOE sample: log real GDP per worker. 62 countries in the file, 7004 observations, 1950-Q1–2024-Q3. Fitted countries: 62. Observations: 7004. Log-likelihood: 18572.70. Christiano–Fitzgerald low-pass trend: periods longer than 100 observations (25 years).

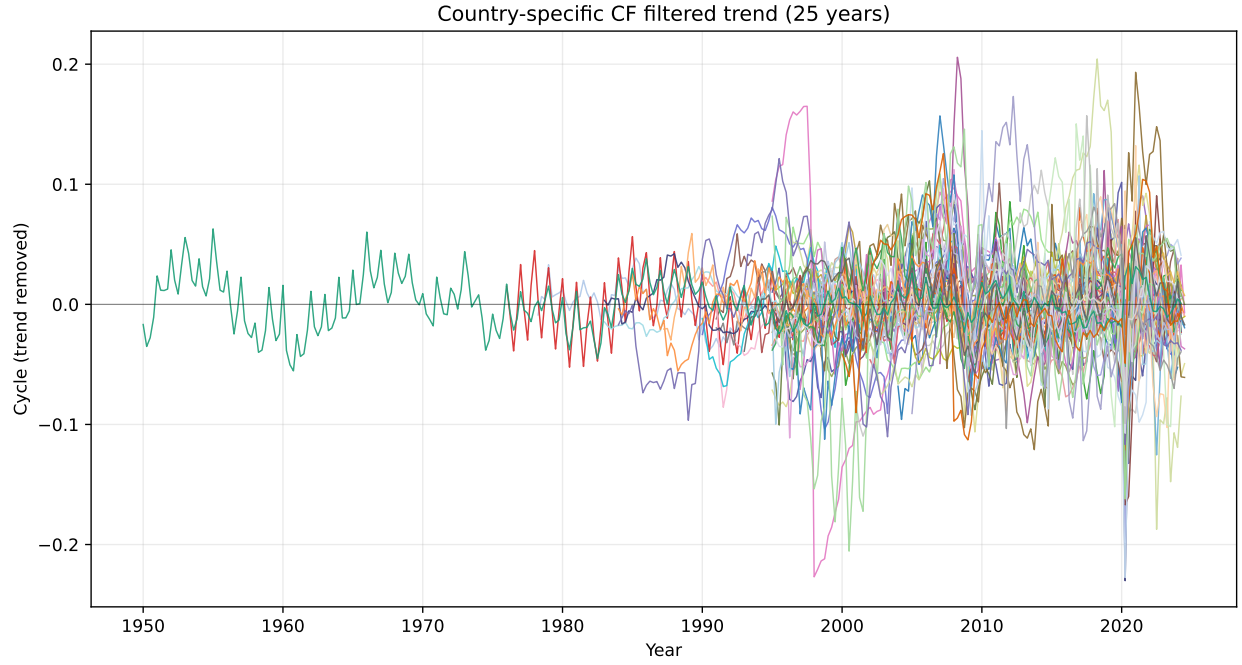


Figure 4: Country cycles after removing the section's trend.

Table 5: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	0.0000 —	0.0000 —	0.0000 —
σ	0.0091 (0.0002)	0.0204 (0.0005)	0.0589 (0.0027)
ρ	0.9264 (0.0091)	0.8441 (0.0155)	0.5308 (0.0501)
$\Pi_0.$	0.9723 (0.0034)	0.0082 (0.0031)	0.0194 (0.0034)
$\Pi_1.$	0.0165 (0.0040)	0.9638 (0.0055)	0.0198 (0.0044)
$\Pi_2.$	0.0681 (0.0191)	0.1613 (0.0325)	0.7706 (0.0322)

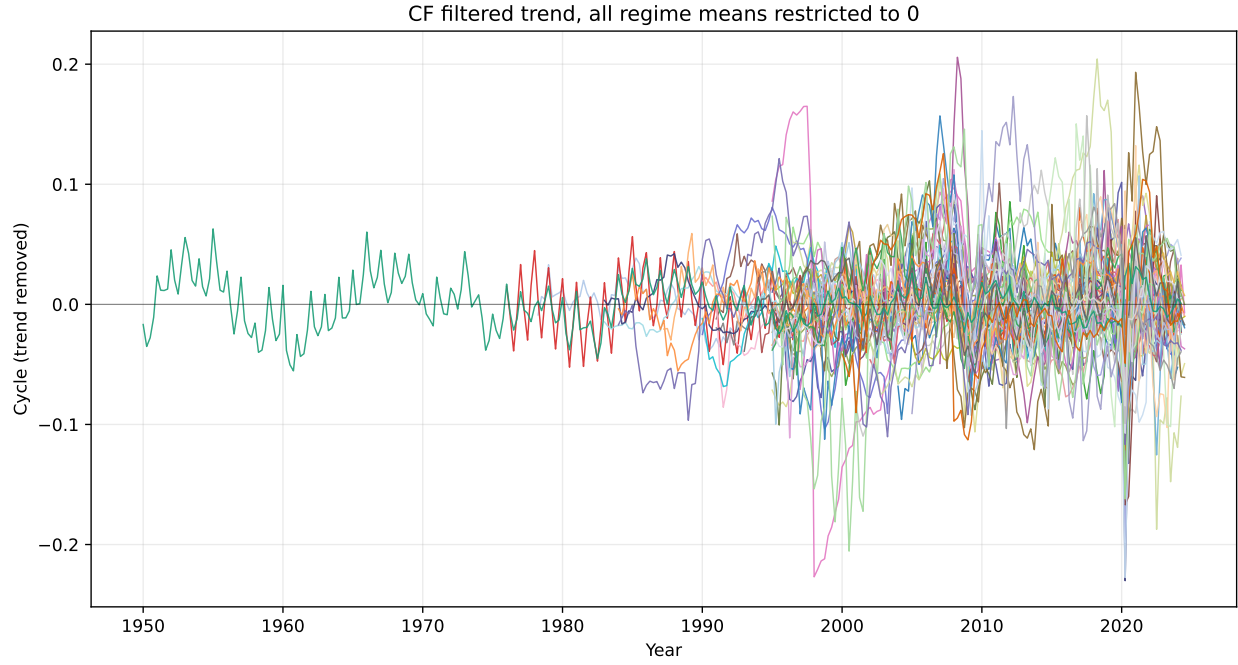


Figure 5: Country cycles after removing the section's trend.

Table 6: Regime AR(1) and transition matrix. Columns are regimes.

	(0)	(1)	(2)
μ	-0.0304 (0.0246)	0.0000 —	0.0017 (0.0032)
σ	0.0656 (0.0030)	0.0093 (0.0002)	0.0208 (0.0005)
ρ (common)	0.8822 (0.0061)	0.8822 (0.0061)	0.8822 (0.0061)
$\Pi_0.$	0.7624 (0.0322)	0.0661 (0.0202)	0.1715 (0.0340)
$\Pi_1.$	0.0172 (0.0030)	0.9742 (0.0034)	0.0086 (0.0032)
$\Pi_2.$	0.0191 (0.0041)	0.0166 (0.0041)	0.9643 (0.0055)

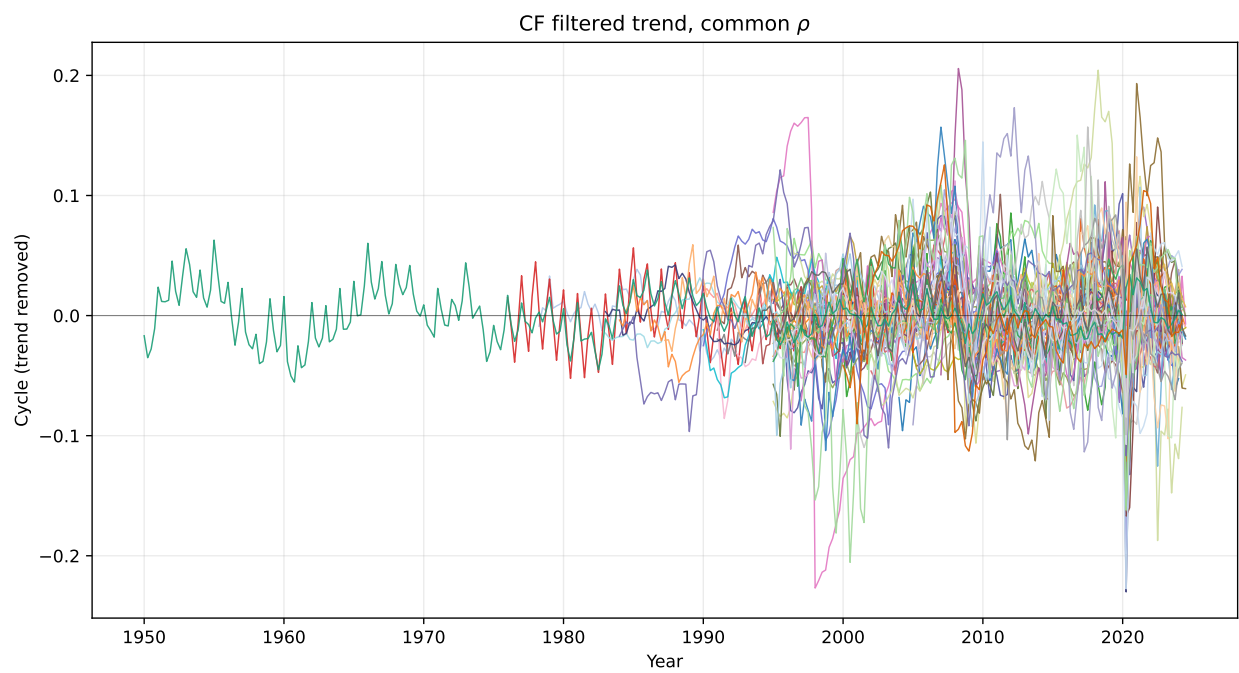


Figure 6: Country cycles after removing the section's trend.